

Algebra and Discrete Mathematics (ADM)

Tutorial 3 Matrix inverse and solving linear systems

Lecturer: doc. Xiaolu Hou, PhD.
xiaolu.hou@stuba.sk

Inverse of 2×2 matrices

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

when

$$\det(A) = ad - bc \neq 0,$$

we have

$$A^{-1} = \frac{1}{\det(A)} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Inverse of 2×2 matrices

$$A = \begin{pmatrix} 2 & 2 \\ 1 & -1 \end{pmatrix}$$

The determinant of A is

$$\det(A) = 2 \times (-1) - 1 \times 2 = -4 \neq 0$$

Then

$$A^{-1} = \frac{1}{\det(A)} \begin{pmatrix} -1 & -2 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} \frac{1}{4} & \frac{1}{2} \\ \frac{1}{4} & -\frac{1}{2} \end{pmatrix}$$

Or by row operations

$$\left(\begin{array}{cc|cc} 2 & 2 & 1 & 0 \\ 1 & -1 & 0 & 1 \end{array} \right) \xrightarrow{R_1 \rightarrow \frac{1}{2}R_1} \left(\begin{array}{cc|cc} 1 & 1 & \frac{1}{2} & 0 \\ 1 & -1 & 0 & 1 \end{array} \right) \xrightarrow{R_2 \rightarrow R_2 - R_1} \left(\begin{array}{cc|cc} 1 & 1 & \frac{1}{2} & 0 \\ 0 & -2 & -\frac{1}{2} & 1 \end{array} \right)$$

Inverse of 2×2 matrices

$$A = \begin{pmatrix} 2 & 2 \\ 1 & -1 \end{pmatrix}$$

By row operations

$$\left(\begin{array}{cc|cc} 1 & 1 & \frac{1}{2} & 0 \\ 0 & -2 & -\frac{1}{2} & 1 \end{array} \right) \xrightarrow{R_2 \rightarrow -\frac{1}{2}R_2} \left(\begin{array}{cc|cc} 1 & 1 & \frac{1}{2} & 0 \\ 0 & 1 & \frac{1}{4} & -\frac{1}{2} \end{array} \right) \xrightarrow{R_1 \rightarrow R_1 - R_2} \left(\begin{array}{cc|cc} 1 & 0 & \frac{1}{4} & \frac{1}{2} \\ 0 & 1 & \frac{1}{4} & -\frac{1}{2} \end{array} \right)$$

Inverse of 2×2 matrices

For $A \in \mathcal{M}_{2 \times 2}$, when do we have $A^{-1} = A$?

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \frac{1}{\det(A)} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

$$a = \frac{d}{\det(A)}, \quad b = -\frac{b}{\det(A)}, \quad c = -\frac{c}{\det(A)}, \quad d = \frac{a}{\det(A)}$$

- If $\det(A) \neq -1$, then $b = c = 0$. Since A is invertible,

$$\det(A) = ad \neq 0.$$

Moreover,

$$a = \frac{d}{ad} = \frac{1}{a}, \quad d = \frac{a}{ad} = \frac{1}{d},$$

so $a^2 = d^2 = 1$. Since $ad = \det(A) \neq -1$, we must have $a = d$. Hence

$$A = I_2 \quad \text{or} \quad A = -I_2.$$

Inverse of 2×2 matrices

For $A \in \mathcal{M}_{2 \times 2}$, when do we have $A^{-1} = A$?

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \frac{1}{\det(A)} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

$$a = \frac{d}{\det(A)}, \quad b = -\frac{b}{\det(A)}, \quad c = -\frac{c}{\det(A)}, \quad d = \frac{a}{\det(A)}$$

- If $\det(A) = -1$, then $a = -d$, so

$$A = \begin{pmatrix} a & b \\ c & -a \end{pmatrix}, \quad a^2 + bc = 1.$$

Inverse of 2×2 matrices

$$A = \begin{pmatrix} 4 & 3 \\ -5 & -4 \end{pmatrix}$$

Then

$$A^{-1} = \frac{1}{4 \times (-4) - 3 \times (-5)} \begin{pmatrix} -4 & -3 \\ 5 & 4 \end{pmatrix} = - \begin{pmatrix} -4 & -3 \\ 5 & 4 \end{pmatrix} = \begin{pmatrix} 4 & 3 \\ -5 & -4 \end{pmatrix} = A$$

Solve linear system using matrix inverse

$$\begin{aligned} -y + z &= 2 \\ 2x + 3y &= 12 \\ -x - z &= -7 \end{aligned}$$

The system can be presented by $A\mathbf{x} = \mathbf{b}$, where

$$A = \begin{pmatrix} 0 & -1 & 1 \\ 2 & 3 & 0 \\ -1 & 0 & -1 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 2 \\ 12 \\ -7 \end{pmatrix}$$

Find inverse of A

$$\left(\begin{array}{ccc|ccc} 0 & -1 & 1 & 1 & 0 & 0 \\ 2 & 3 & 0 & 0 & 1 & 0 \\ -1 & 0 & -1 & 0 & 0 & 1 \end{array} \right) \xrightarrow{(-R_3) \leftrightarrow R_1} \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 1 & 0 \\ 0 & -1 & 1 & 1 & 0 & 0 \end{array} \right)$$

Solve linear system using matrix inverse

$$\begin{aligned} -y + z &= 2 \\ 2x + 3y &= 12 \\ -x - z &= -7 \end{aligned}$$

Find inverse of A

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 1 & 0 \\ 0 & -1 & 1 & 1 & 0 & 0 \end{array} \right) \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 0 & 0 & -1 \\ 0 & 3 & -2 & 0 & 1 & 2 \\ 0 & -1 & 1 & 1 & 0 & 0 \end{array} \right) \xrightarrow[R_2 \leftrightarrow (-R_3)]{R_2 \rightarrow R_2 + 3R_3}$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 0 & 0 & -1 \\ 0 & 1 & -1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 3 & 1 & 2 \end{array} \right) \xrightarrow[R_1 \rightarrow R_1 - R_3]{R_2 \rightarrow R_2 + R_3} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -3 & -1 & -3 \\ 0 & 1 & 0 & 2 & 1 & 2 \\ 0 & 0 & 1 & 3 & 1 & 2 \end{array} \right)$$

Solve linear system using matrix inverse

$$-y + z = 2$$

$$2x + 3y = 12$$

$$-x - z = -7$$

$$A^{-1} = \begin{pmatrix} -3 & -1 & -3 \\ 2 & 1 & 2 \\ 3 & 1 & 2 \end{pmatrix}$$

The solution is given by

$$\mathbf{x} = A^{-1}\mathbf{b} = \begin{pmatrix} -3 & -1 & -3 \\ 2 & 1 & 2 \\ 3 & 1 & 2 \end{pmatrix} \begin{pmatrix} 2 \\ 12 \\ -7 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ 4 \end{pmatrix}$$

Solve linear system using matrix inverse

$$A = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 2 & 1 & 1 & 1 \\ 1 & 1 & 0 & -1 \\ 1 & 2 & 2 & 0 \end{pmatrix}, \quad b_1 = \begin{pmatrix} 2 \\ 4 \\ -2 \\ 7 \end{pmatrix}, \quad b_2 = \begin{pmatrix} 4 \\ 8 \\ -7 \\ 8 \end{pmatrix}$$

$$\left(\begin{array}{cccc|cccc} 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & -1 & 0 & 0 & 1 & 0 \\ 1 & 2 & 2 & 0 & 0 & 0 & 0 & 1 \end{array} \right) \xrightarrow[\substack{R_3 \rightarrow R_3 - R_1, \quad R_4 \rightarrow R_4 - R_1}]{R_2 \rightarrow R_2 - 2R_1} \left(\begin{array}{cccc|cccc} 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & -1 & -2 & 1 & 0 & 0 \\ 0 & 1 & -1 & -2 & -1 & 0 & 1 & 0 \\ 0 & 2 & 1 & -1 & -1 & 0 & 0 & 1 \end{array} \right)$$

$$\xrightarrow[\substack{R_4 \rightarrow R_4 - 2R_2}]{R_3 \rightarrow R_3 - R_2} \left(\begin{array}{cccc|cccc} 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & -1 & -2 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 3 & 1 & 3 & -2 & 0 & 1 \end{array} \right)$$

Solve linear system using matrix inverse

$$\left(\begin{array}{cccc|cccc} 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & -1 & -2 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 3 & 1 & 3 & -2 & 0 & 1 \end{array} \right) \xrightarrow{\frac{1}{3}R_4 \leftrightarrow (-R_3)} \left(\begin{array}{cccc|cccc} 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & -1 & -2 & 1 & 0 & 0 \\ 0 & 0 & 1 & \frac{1}{3} & 1 & -\frac{2}{3} & 0 & \frac{1}{3} \\ 0 & 0 & 0 & 1 & -1 & 1 & -1 & 0 \end{array} \right)$$

$$\xrightarrow[\begin{array}{l} R_2 \rightarrow R_2 + R_3 \\ R_1 \rightarrow R_1 - R_3 \end{array}]{\phantom{\frac{1}{3}R_4 \leftrightarrow (-R_3)}} \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & \frac{2}{3} & 0 & \frac{2}{3} & 0 & -\frac{1}{3} \\ 0 & 1 & 0 & -\frac{2}{3} & -1 & \frac{1}{3} & 0 & \frac{1}{3} \\ 0 & 0 & 1 & \frac{1}{3} & 1 & -\frac{2}{3} & 0 & \frac{1}{3} \\ 0 & 0 & 0 & 1 & -1 & 1 & -1 & 0 \end{array} \right)$$

Solve linear system using matrix inverse

$$\left(\begin{array}{cccc|cccc} 1 & 0 & 0 & \frac{2}{3} & 0 & \frac{2}{3} & 0 & -\frac{1}{3} \\ 0 & 1 & 0 & -\frac{2}{3} & -1 & \frac{1}{3} & 0 & \frac{1}{3} \\ 0 & 0 & 1 & \frac{1}{3} & 1 & -\frac{2}{3} & 0 & \frac{1}{3} \\ 0 & 0 & 0 & 1 & -1 & 1 & -1 & 0 \end{array} \right) \xrightarrow[\substack{R_2 \rightarrow R_2 + \frac{2}{3}R_4, \quad R_3 \rightarrow R_3 - \frac{1}{3}R_4}]{R_1 \rightarrow R_1 - \frac{2}{3}R_4}$$

$$\left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & \frac{2}{3} & 0 & \frac{2}{3} & -\frac{1}{3} \\ 0 & 1 & 0 & 0 & -\frac{5}{3} & 1 & -\frac{2}{3} & \frac{1}{3} \\ 0 & 0 & 1 & 0 & \frac{4}{3} & -1 & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & 0 & 1 & -1 & 1 & -1 & 0 \end{array} \right) \Rightarrow A^{-1} = \frac{1}{3} \begin{pmatrix} 2 & 0 & 2 & -1 \\ -5 & 3 & -2 & 1 \\ 4 & -3 & 1 & 1 \\ -3 & 3 & -3 & 0 \end{pmatrix}$$

Solve linear system using matrix inverse

$$A = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 2 & 1 & 1 & 1 \\ 1 & 1 & 0 & -1 \\ 1 & 2 & 2 & 0 \end{pmatrix}, \quad \mathbf{b}_1 = \begin{pmatrix} 2 \\ 4 \\ -2 \\ 7 \end{pmatrix}, \quad \mathbf{b}_2 = \begin{pmatrix} 4 \\ 8 \\ -7 \\ 8 \end{pmatrix}$$

$$\mathbf{x}_1 = A^{-1}\mathbf{b}_1 = \frac{1}{3} \begin{pmatrix} 2 & 0 & 2 & -1 \\ -5 & 3 & -2 & 1 \\ 4 & -3 & 1 & 1 \\ -3 & 3 & -3 & 0 \end{pmatrix} \begin{pmatrix} 2 \\ 4 \\ -2 \\ 7 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} -7 \\ 13 \\ 1 \\ 12 \end{pmatrix}$$

$$\mathbf{x}_2 = A^{-1}\mathbf{b}_2 = \frac{1}{3} \begin{pmatrix} 2 & 0 & 2 & -1 \\ -5 & 3 & -2 & 1 \\ 4 & -3 & 1 & 1 \\ -3 & 3 & -3 & 0 \end{pmatrix} \begin{pmatrix} 4 \\ 8 \\ -7 \\ 8 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} -14 \\ 26 \\ -7 \\ 33 \end{pmatrix}$$

Solve linear system using matrix inverse

$$A = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 2 & 1 & 1 & 1 \\ 1 & 1 & 0 & -1 \\ 1 & 2 & 2 & 0 \end{pmatrix}, \quad \mathbf{b}_1 = \begin{pmatrix} 2 \\ 4 \\ -2 \\ 7 \end{pmatrix}, \quad \mathbf{b}_2 = \begin{pmatrix} 4 \\ 8 \\ -7 \\ 8 \end{pmatrix}$$

We can verify that

$$A\mathbf{x}_1 = \frac{1}{3} \begin{pmatrix} 1 & 0 & 1 & 1 \\ 2 & 1 & 1 & 1 \\ 1 & 1 & 0 & -1 \\ 1 & 2 & 2 & 0 \end{pmatrix} \begin{pmatrix} -7 \\ 13 \\ 1 \\ 12 \end{pmatrix} = \mathbf{b}_1$$

$$A\mathbf{x}_2 = \frac{1}{3} \begin{pmatrix} 1 & 0 & 1 & 1 \\ 2 & 1 & 1 & 1 \\ 1 & 1 & 0 & -1 \\ 1 & 2 & 2 & 0 \end{pmatrix} \begin{pmatrix} -14 \\ 26 \\ -7 \\ 33 \end{pmatrix} = \mathbf{b}_2$$

Update to Homework Assessment Criteria

To give you another opportunity to improve your homework marks, the following changes will apply:

- **Resubmission opportunity for T1–T3:** You may revise and resubmit your solutions to Tutorials 1–3.
- **Deadline:** 15 October 2026, 10am
- **Updated marking scheme:** You can earn up to **24 marks** in total across all homework assignments, including **14 extra marks**.

Please use this opportunity to complete missing questions, correct errors, and improve the clarity of your solutions.

Homework Expectations and Resubmission Policy (T4–T12)

- **Make a genuine effort** to solve all required homework questions before each tutorial.
- You are expected to understand the methods and reasoning, not simply copy the provided solutions.
- **Grading criteria:**
 - **2 marks:** All required questions are meaningfully attempted, with correct and sufficiently detailed solutions.
 - **1 mark:** All required questions are meaningfully attempted, but some solutions contain errors or incomplete reasoning.
 - **0 marks:** One or more required questions are not meaningfully attempted.
- **Resubmission opportunity (T4–T12 only):** If you receive **1 mark**, you may revise and resubmit your homework after the tutorial to improve your grade to **2 marks**.
- **Resubmission deadline:** You may resubmit your homework either in person or via UIS. Strictly follow the deadline specified in UIS; **no late submissions will be accepted**.

Meaningful attempts

- **If you can solve the question:** Show your calculations and mathematical steps as usual. No additional written explanations are necessary.
- **If you cannot solve the question:** Follow the approach illustrated in the next slides to demonstrate a meaningful attempt.

Example of a Meaningful Attempt: Q24(1) (1/3)

Given

$$A = \begin{pmatrix} -3 & 1 \\ 2 & 2 \end{pmatrix}.$$

We apply the matrix inversion algorithm:

$$\left(\begin{array}{cc|cc} -3 & 1 & 1 & 0 \\ 2 & 2 & 0 & 1 \end{array} \right) \xrightarrow{R_1 \rightarrow -\frac{1}{3}R_1} \left(\begin{array}{cc|cc} 1 & -\frac{1}{3} & -\frac{1}{3} & 0 \\ 2 & 2 & 0 & 1 \end{array} \right)$$

$$\xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left(\begin{array}{cc|cc} 1 & -\frac{1}{3} & -\frac{1}{3} & 0 \\ 0 & \frac{8}{3} & \frac{2}{3} & 1 \end{array} \right)$$

$$\xrightarrow{R_2 \rightarrow \frac{3}{8}R_2} \left(\begin{array}{cc|cc} 1 & -\frac{1}{3} & -\frac{1}{3} & 0 \\ 0 & 1 & \frac{1}{4} & \frac{3}{8} \end{array} \right)$$

$$\xrightarrow{R_1 \rightarrow R_1 + \frac{1}{3}R_2} \left(\begin{array}{cc|cc} 1 & 0 & -\frac{1}{4} & \frac{1}{8} \\ 0 & 1 & \frac{1}{4} & \frac{3}{8} \end{array} \right).$$

Example of a Meaningful Attempt: Q24(1) (2/3)

The corresponding elementary matrices are

$$E_1 = \begin{pmatrix} -\frac{1}{3} & 0 \\ 0 & 1 \end{pmatrix}, \quad E_2 = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix},$$

$$E_3 = \begin{pmatrix} 1 & 0 \\ 0 & \frac{3}{8} \end{pmatrix}, \quad E_4 = \begin{pmatrix} 1 & \frac{1}{3} \\ 0 & 1 \end{pmatrix}.$$

Their inverses are

$$E_1^{-1} = \begin{pmatrix} -3 & 0 \\ 0 & 1 \end{pmatrix}, \quad E_2^{-1} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix},$$

$$E_3^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & \frac{8}{3} \end{pmatrix}, \quad E_4^{-1} = \begin{pmatrix} 1 & -\frac{1}{3} \\ 0 & 1 \end{pmatrix}.$$

We have

$$E_4 E_3 E_2 E_1 A = I.$$

Therefore,

$$A^{-1} = E_4 E_3 E_2 E_1, \quad A = E_1^{-1} E_2^{-1} E_3^{-1} E_4^{-1}.$$

Example of a Meaningful Attempt: Q24(1) (3/3)

- **First row operation:** I understand that multiplying R_1 by $-\frac{1}{3}$ makes the first pivot equal to 1.
- **Second row operation:** We subtract twice the first row from the second row to eliminate the entry below the first pivot.
- **Third and fourth row operations:** I understand that these operations make the second pivot equal to 1 and eliminate the entry above it.
- **Elementary matrices:** I understand that each elementary matrix represents one row operation applied to the identity matrix.
- **What I do not understand:** I do not understand why

$$E_4 E_3 E_2 E_1 A = I$$

implies

$$A = E_1^{-1} E_2^{-1} E_3^{-1} E_4^{-1}.$$

In particular, why is the order reversed?

Meaningful Attempt: Q24(2)

Given

$$A = \begin{pmatrix} 1 & 0 \\ -5 & 2 \end{pmatrix}.$$

I learned from the worked solution to Part 1 that we should use row operations to transform A into I .

First, I apply $R_2 \rightarrow R_2 + 5R_1$:

$$\begin{pmatrix} 1 & 0 \\ -5 & 2 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}.$$

Then, I apply $R_2 \rightarrow \frac{1}{2}R_2$:

$$\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I.$$

The corresponding elementary matrices are

$$E_1 = \begin{pmatrix} 1 & 0 \\ 5 & 1 \end{pmatrix}, \quad E_2 = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}.$$

Where I get stuck: I know that $E_2 E_1 A = I$, but I am not sure how to use this to express A and A^{-1} as products of elementary matrices.

Meaningful Attempt: Q24(2) – Another Example

Given

$$A = \begin{pmatrix} 1 & 0 \\ -5 & 2 \end{pmatrix}.$$

I learned from the worked solution to Part 1 that we should use row operations to transform A into I .

First, I apply $R_2 \rightarrow R_2 + 5R_1$:

$$\begin{pmatrix} 1 & 0 \\ -5 & 2 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}.$$

Then, I apply $R_2 \rightarrow \frac{1}{2}R_2$:

$$\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I.$$

Where I get stuck:

In the worked solution to Part 1, each row operation has a corresponding elementary matrix E_i . However, I do not understand how to construct these elementary matrices from the row operations. Therefore, I cannot continue to express A and A^{-1} as products of elementary matrices.

Meaningful Attempt: Question 21 (1/6)

Example of a student's submission

Question 21: Use the inversion algorithm to find the inverse of the matrix (if the inverse exists). Choose any two subparts.

What I do not understand:

I do not understand how to use the inversion algorithm to calculate the inverse of a matrix.

What I did:

I went back to the lecture slides and found three slides related to this method:

- 1 *Inversion algorithm*
- 2 *Inversion algorithm – example*
- 3 *Inversion algorithm – example (continued)*

I copied the relevant method and worked example, then tried to understand each step before applying the method to Question 21.

Meaningful Attempt: Question 21 (2/6)

My notes from the lecture: Inversion algorithm

The lecture explains that we should:

- Find elementary row operations that reduce A to I_n .
- Perform the same operations on I_n .
- Start with the augmented matrix

$$\left(A \mid I_n \right)$$

and transform it into

$$\left(I_n \mid A^{-1} \right).$$

My understanding:

I understand that we want to transform the left-hand side into the identity matrix. The right-hand side will then give us the inverse.

However, I am not always sure how to choose the appropriate row operations.

Meaningful Attempt: Question 21 (3/6)

Copied from the lecture: Inversion algorithm – example

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{pmatrix}$$

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array} \right) \xrightarrow[R_3 \rightarrow R_3 - R_1]{R_2 \rightarrow R_2 - 2R_1} \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & -2 & 5 & -1 & 0 & 1 \end{array} \right)$$

$$\xrightarrow{R_3 \rightarrow R_3 + 2R_2} \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & -1 & -5 & 2 & 1 \end{array} \right)$$

$$\xrightarrow{R_3 \rightarrow -R_3} \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right)$$

My understanding:

- The first two operations eliminate the entries below the first pivot.
- The next operation eliminates the entry below the second pivot.
- Multiplying the third row by -1 makes the third pivot equal to 1.

Meaningful Attempt: Question 21 (4/6)

Copied from the lecture: Example (continued)

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right) \xrightarrow[R_2 \rightarrow R_2 + 3R_3]{R_1 \rightarrow R_1 - 3R_3} \left(\begin{array}{ccc|ccc} 1 & 2 & 0 & -14 & 6 & 3 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right)$$
$$\xrightarrow{R_1 \rightarrow R_1 - 2R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -40 & 16 & 9 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right)$$

Thus,

$$A^{-1} = \begin{pmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{pmatrix}.$$

My understanding:

The last operations eliminate the entries above the pivots, giving the identity matrix on the left.

What I still do not understand:

I can follow these operations, but I find it difficult to decide which row operation to use next when solving a new problem independently.

Meaningful Attempt: Question 21 (5/6)

My attempt at Question 21, Part 1

Given

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix}.$$

Following the lecture example, I start with

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \end{array} \right).$$

I can eliminate the entry below the first pivot using $R_3 \rightarrow R_3 - R_1$:

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & -1 & 0 & 1 \end{array} \right).$$

Where I get stuck:

In the lecture example, the next operation eliminates the entry below the second pivot. I think I need to do something similar here, but I am not sure how to choose the next operation and continue until the left-hand side becomes I_3 .

Meaningful Attempt: Question 21 (6/6)

My attempt at Question 21, Part 3

Given

$$A = \begin{pmatrix} 2 & 6 & 6 \\ 2 & 7 & 6 \\ 2 & 7 & 7 \end{pmatrix}.$$

I begin with $[A \mid I_3]$ and apply $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$\left(\begin{array}{ccc|ccc} 2 & 6 & 6 & 1 & 0 & 0 \\ 2 & 7 & 6 & 0 & 1 & 0 \\ 2 & 7 & 7 & 0 & 0 & 1 \end{array} \right) \longrightarrow \left(\begin{array}{ccc|ccc} 2 & 6 & 6 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 1 & 1 & -1 & 0 & 1 \end{array} \right).$$

Then I use $R_3 \rightarrow R_3 - R_2$:

$$\left(\begin{array}{ccc|ccc} 2 & 6 & 6 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{array} \right).$$

Where I get stuck:

I understand how to eliminate the entries below the pivots. However, I am not sure in which order to eliminate the entries above the pivots and transform the left-hand side into I_3 .

This is my attempt at both required subparts.

Quiz one information

- Time: 9:30–11:50
- Location: same room as the lecture
- Toilet breaks are not permitted during the quiz.
- Calculators are not allowed.
- Write your answers on the provided answer sheets. Additional sheets will be supplied upon request.
- Write your full name clearly on every page of your answer sheets.
- Show all computation steps in sufficient detail. Answers without supporting calculations will receive zero points, even if the final answer is correct.

Quiz one information

- Q1: dot product, orthogonality, and vector norm
- Q2: matrix multiplication, transpose, scalar multiplication, and matrix subtraction
- Q3: linear systems, Gauss–Jordan elimination, consistency, and solution sets
- Q4: matrix inversion using row operations, elementary matrices and their inverses, and products of elementary matrices